Quiz Q1. Let

$$\mathbf{A} = \begin{pmatrix} 9 & -2 \\ -2 & 6 \end{pmatrix}, \mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ -3 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}, \mathbf{v}_4 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Which option lists exactly the eigenvectors of $\mathbf{A}$ among these four vectors?

A. $\mathbf{v}_1, \mathbf{v}_4$

B. $\mathbf{v}_2, \mathbf{v}_3$

C. $\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_4$

D. $\mathbf{v}_1, \mathbf{v}_3$

Quiz Q1 answer. The correct answer is D.

The two nonzero eigenvectors satisfy

$$\mathbf{A}\mathbf{v}_1 = \begin{pmatrix} 5 \\ 10 \end{pmatrix} = 5\mathbf{v}_1, \qquad \mathbf{A}\mathbf{v}_3 = \begin{pmatrix} -20 \\ 10 \end{pmatrix} = 10\mathbf{v}_3.$$

$$\mathbf{A}\mathbf{v}_2 = \begin{pmatrix} 15 \\ -20 \end{pmatrix}.$$

For $\mathbf{v}_2$, the coordinate ratios $15/1 = 15$ and $(-20)/(-3) = 20/3$ differ. Thus $\mathbf{v}_2$ is not an eigenvector.

The zero vector $\mathbf{v}_4$ is never an eigenvector.

Quiz Q2. Let

$$\mathbf{C}_1 = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}, \mathbf{C}_2 = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}, \mathbf{C}_3 = \begin{pmatrix} 3 & -1 \\ 1 & 2 \end{pmatrix}.$$

Which option lists exactly the matrices that cannot be a dataset's covariance matrix?

A. $\mathbf{C}_1$

B. $\mathbf{C}_3$

C. $\mathbf{C}_1, \mathbf{C}_3$

D. $\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3$

Quiz Q2 answer. The correct answer is C.

Covariance matrices are symmetric and positive semidefinite.

$\mathbf{C}_1$ has eigenvalue -1, with eigenvector $(1, -1)^\mathsf{T}$, so it is not positive semidefinite.

$\mathbf{C}_3$ is not symmetric: its off-diagonal entries differ.

$\mathbf{C}_2$ is symmetric with eigenvalues 5 and 1, so it is a valid covariance matrix. For example, it is the covariance of the four centered datapoints $\pm(\sqrt{5}, \sqrt{5})^\mathsf{T}$ and $\pm(1, -1)^\mathsf{T}$ under our $1/n$ convention.

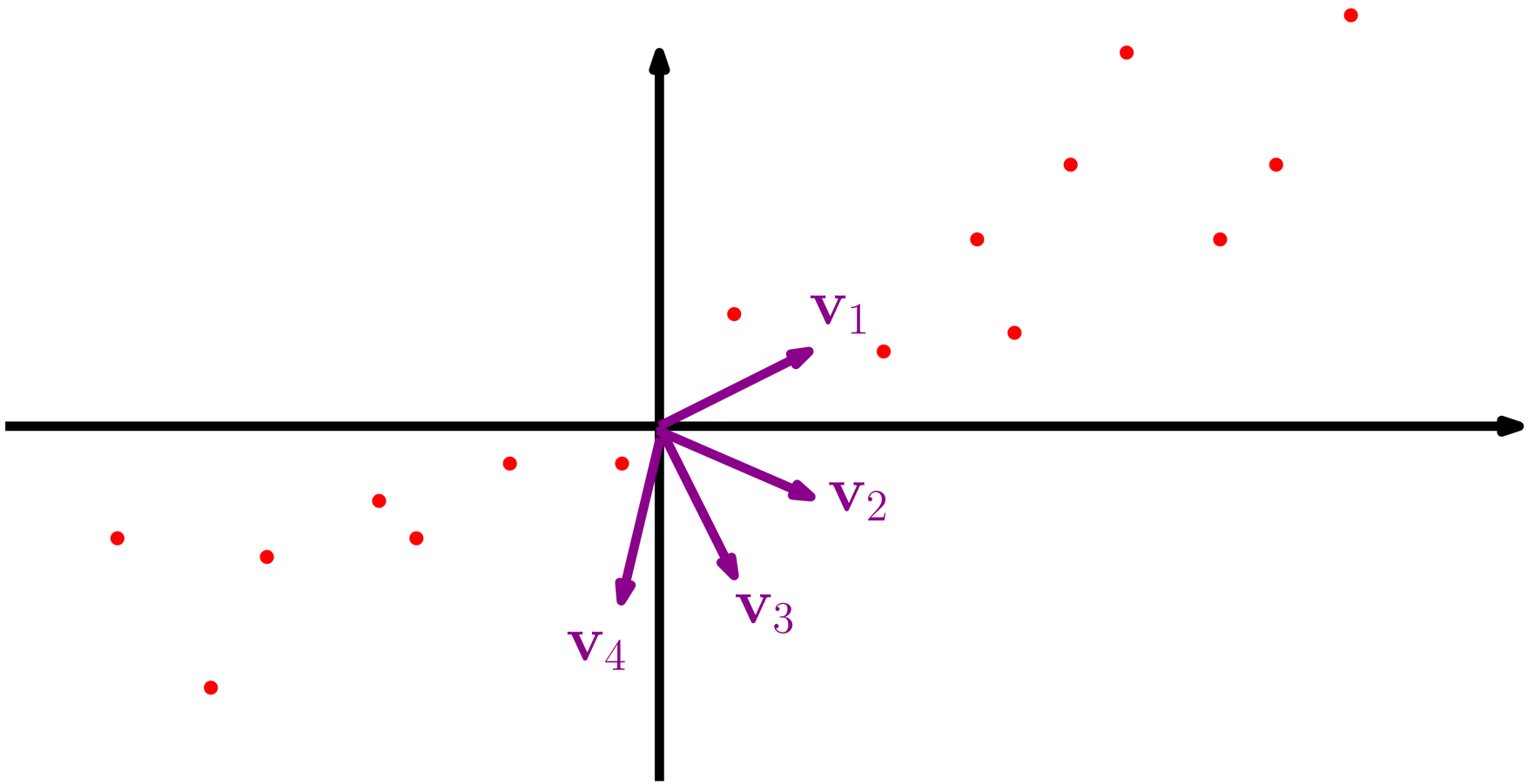
Quiz Q3. Which direction gives the largest variance of the plotted datapoints' orthogonal projections onto its line?

A. $\mathbf{v}_1$

B. $\mathbf{v}_2$

C. $\mathbf{v}_3$

D. $\mathbf{v}_4$



Quiz Q3 answer. The correct answer is A.

The cloud extends mainly along $\mathbf{v}_1$. Its orthogonal projections have the largest spread among the four choices.

A nearly perpendicular direction, such as $\mathbf{v}_3$, compresses this spread.

Quiz Q1. PCA uses the directions

$$\mathbf{q}_1 = \frac{1}{2}\begin{pmatrix}1\\1\\1\\1\end{pmatrix}, \qquad \mathbf{q}_2 = \frac{1}{2}\begin{pmatrix}1\\1\\-1\\-1\end{pmatrix}.$$

A new datapoint has $\mathbf{x} - \boldsymbol{\mu} = (12, 8, -4, -4)^\mathsf{T}$. What are its two retained PCA scores?

A. $(12, 8)^\mathsf{T}$

B. $(6, -14)^\mathsf{T}$

C. $(6, 14)^\mathsf{T}$

D. $(3, 7)^\mathsf{T}$

Quiz Q1 answer. The correct answer is C.

Compute the two inner products:

$$\langle \mathbf{q}_1, \mathbf{x} - \boldsymbol{\mu} \rangle = \frac{12 + 8 - 4 - 4}{2} = 6,$$

$$\langle \mathbf{q}_2, \mathbf{x} - \boldsymbol{\mu} \rangle = \frac{12 + 8 + 4 + 4}{2} = 14.$$

Thus $\mathbf{z} = (6, 14)^\mathsf{T}$.

PCA scores are inner products with the unit PC directions, rather than selected original feature values.

Quiz Q2. A covariance matrix has ordered eigenvalues

$$35,\quad 31,\quad 25,\quad 3,\quad 2,\quad 1.5,\quad 1,\quad 1,\quad 0.5$$

Their sum is 100. What are the smallest numbers of PCs needed to retain at least 90% and at least 95% of the variance, respectively?

A. 4 and 6

B. 3 and 5

C. 3 and 6

D. 5 and 7

Quiz Q2 answer. The correct answer is B.

The total variance is 100. The cumulative retained fractions are

$$\rho_2 = \frac{35+31}{100} = 66\% < 90\%,$$
$$\rho_3 = \frac{35+31+25}{100} = 91\% \geq 90\%,$$
$$\rho_4 = \frac{91+3}{100} = 94\% < 95\%,$$
$$\rho_5 = \frac{94+2}{100} = 96\% \geq 95\%.$$

Thus the smallest numbers are 3 and 5, respectively.

Quiz Q3. Consider a dataset with eight datapoints $(\pm 20, \pm 10, \pm 3)^{\mathsf{T}}$, with all sign combinations included.

The label is $+1$ when the third feature is positive and -1 otherwise.

Apply PCA without feature scaling and retain the first two PCs. Which statement is correct?

A. Retaining more than 98% of variance guarantees that the labels remain distinguishable.

B. Datapoints with opposite labels can overlap in the retained representation.

C. The third feature has more variance than the first.

D. PCA uses the labels to retain the third feature.

Quiz Q3 answer. The correct answer is B.

Symmetry gives zero mean and zero cross-covariances:

$$\mathbf{C} \;=\; \mathrm{diag}(400, 100, 9).$$

The leading PCs are the first two coordinate axes, up to sign:

$$\rho_2 \;=\; \frac{400 + 100}{400 + 100 + 9} \;\approx\; 98.23\%.$$

The points $(20, 10, 3)^{\mathsf{T}}$ and $(20, 10, -3)^{\mathsf{T}}$ have opposite labels but identical retained scores.

PCA ignores labels and discards the feature defining the classes.